



Lithodat Product and Services

Discussion Forum - 2/09/2025

Keith Dimech

Website: <https://lithodraft.netlify.app/>



Defining Products and Services

Products

- Tangible (or at least standardised and repeatable).
- Defined features, specs, and deliverables.
- Scalable — once designed, can be sold many times with low marginal cost.
- Products can scale quickly

Example - Lithosurfer: the product is providing access to Lithodat owned data through a platform.



Services

- Intangible, process-based, value from man hours, often tailored.
- Created and curated and consumed by the customer.
- Depend on human expertise and interaction
- Services are slow by nature, linked with training and people management

Example - Isotopes.au: consulting on data workflows; understanding and learning customers wants; custom platform development.



Combining Products and Services



A deliberate bundle where a tangible or repeatable products is paired with a tailored services to create a more complete solution for customers.

Examples:

Lithosurfer (Product) + Bespoke Software Tool Build (Service)

Lithosurfer (Product) + Data Mining of public data (Service)

Tesla (Product) + Charging Stations (Product)

When it works well:

- Customers prefer outcomes, not just tools. Bundles solve problems end-to-end.
- Services lock customers into your ecosystem, raising switching costs.
- Products provide upfront sales; services (like subscriptions) generate long-term income.
- Harder for competitors to copy the combination of your product + expertise.
- Services give insights into how customers use products, guiding product improvement.

Combining Products and Services



A deliberate bundle where a tangible or repeatable product is paired with a tailored service to create a more complete solution for customers.

Examples:

Apple iPod (Products) and iTunes (Service)

Hairdresser (Service) + Shampoo (Products)

Lesson: You need to understand *where the value lies* in the each product and service - and charge correctly.

When it works can kill:

- Managing both product and service operations requires different capabilities, staff, and culture.
- If bundles aren't well designed or priced correctly - customers may see them as confusing.
- Services are labour-intensive and can erode the high-margin scalability of products.
- Products scale quickly; services scale slowly. Mixing them can cap growth if not structured carefully.
- Failure in one part of the bundle (e.g. weak service delivery) can damage the reputation of the whole

Services: Reols Framework for Co-Productivity

A framework that maps services by:

- Effort: Who does the work (Provider vs. Customer vs. Shared).
- Site: Where the work is situated (Provider vs. Customer vs. Shared).

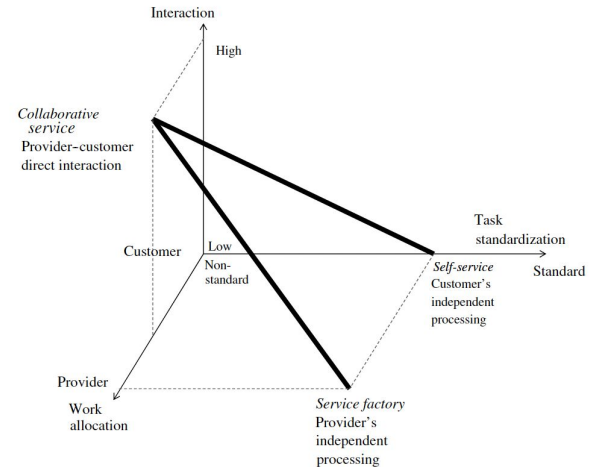
Why it matters:

- Clarifies who creates value and who controls resources.
- Makes visible the trade-offs between scalability, customisation, and cost.
- Highlights the co-production nature of services

Lithodat's products/services range across the matrix:

- Provider-heavy: Data mining, software improvement and standardisation, adaptors.
- Joint effort: Platform development, tool development, consulting.
- Customer-heavy: Lithosurfer, Field apps, client-managed data.

A Product-Process Matrix for Coproductive Services



Lithodats Framework.

Lithodat is unique as we also need to consider the data ownership. This is a third important pillar to our services and products, and with it come additional costs and considerations.

To define this - I have adapted Reols and put all our products through a lens of:

- Who does the work (Lithodat vs customer),
- Where it happens (Lithdat vs customer site),
- Whose resources (data) is used (provider vs customer).

Customer Work			
Joint Work			
Lithodat Work			
	Customer Site	Joint / Shared	Lithodat Site



Our Products and Services

LithoClean™

CLEAN is Lithodat's core data cleaning service. It takes raw, unstructured geochemistry datasets — often scattered across spreadsheets, reports, or lab files — and systematically standardised them. The process involves:

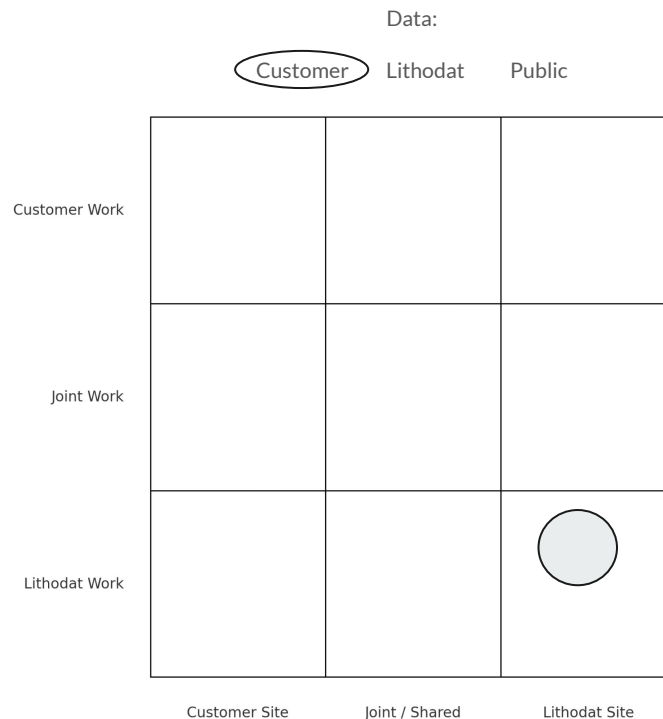
- Identifying inconsistencies in fields, units, naming, and metadata.
- Converting them into a structured, relational database format.

Classification: Service

CLEAN is a service-only activity: Lithodat's experts and processes deliver a result based on the customer's unique raw data. The output is a data export, not a product or platform subscription.

Lithodat does (most) of the work, in our systems, in our offices.

Customer Data: We could clean up clients data (BHP) has historic data.



LithoBuild™

BUILD is Lithodat's platform development service. It focuses on designing and delivering bespoke geoscience data infrastructures for clients. This can mean:

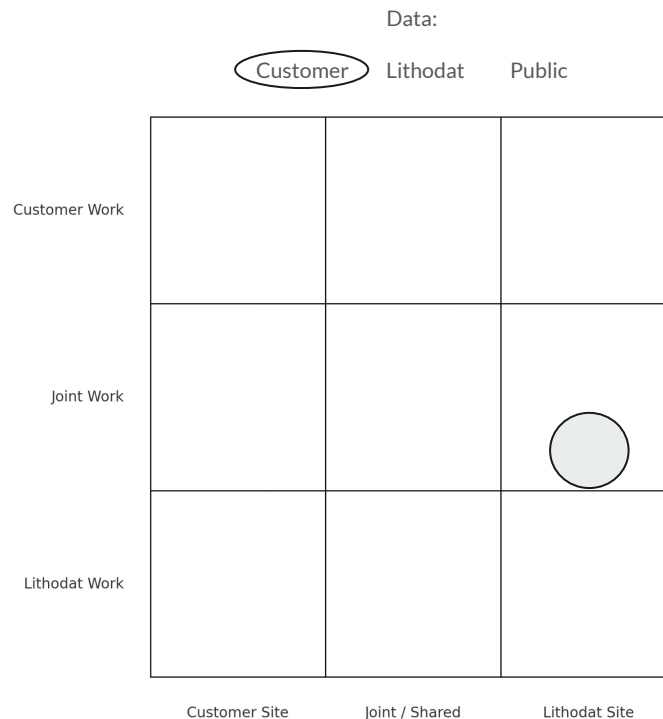
- Creating new data models for specific scientific domains (e.g. isotopes, thermochronology).
- Building adaptors and tools inside Lithosurfer requested by customers.
- Developing whole new platforms or extending existing ones for government surveys, academia, or industry. (ISOtopes)

Classification: Service

Lithodat's experts and processes deliver a result based on the customer's unique raw data. The output may be a product, but BUILD is specific to the service we provide.

Mostly Lithodat Site, Predominantly Lithodat Work - however these platforms will require design input and data from the customer.

Client IP or access to their server will sometimes be needed in order to develop the Build™ service. Specific adaptors or APIs from clients workspace.



LithoMine™

MINE™ is Lithodat's public data acquisition service. When a client, like BHP, wants all the available information for a specific region or domain, Lithodat is tasked with going into the public domain (academic journals, government repositories, survey reports) and systematically extracting it. The process involves:

- Searching across diverse public and semi-public sources.
- Extracting relevant datasets and metadata.
- Delivering the results as a structured, cleaned dataset ready for use.

Classification: Service

MINE is a service-only activity: Lithodat's team uses its technology, resources, and expertise to source and assemble data on behalf of the client. The output is a consolidated dataset, not a repeatable product or subscription. Must be paired with other products/services to be effective.

This work does not involve the client, it completely a lithodat run service on our platform.

This is "Public" data we are mining, however once it has been collected it will be passed onto the customer that paid for it. Beyond this though - we can resell it meaning that this process potentially makes this data 'Lithodats' data for future products.

Data:

Customer

Lithodat

Public

Customer Work

Joint Work

Lithodat Work

Customer Site

Joint / Shared

Lithodat Site



Combining Products and Services



Clean™ + Build™ = When we clean up customers data and then build them a platform or tools to view.

Mine™ + Clean™ = We find the data through Mine then Clean before transferring it to the Customer



LithoData™ - Free

LithoData™ is Lithodat's Data offering and is the most valuable part of our products. The LithoData we have:

LithoData Free

- AGN data
- Any future research collaboration
- Any future grant or public government data (isotopes, Saudi etc)
- Any data that any users have made available to the public.

Classification: Product

LithoData Free is available through a LithoSurfer login. All data is delivered through the LithoSurfer platform. The customer must have at least a free Lithosurfer account.

		Data:			
		Customer	Lithodat	Public	
Customer Work	Joint Work				
Lithodat Work					
		Customer Site	Joint / Shared	Lithodat Site	



LithoData™ - Platinum

LithoData™ Premium is Lithodat's Data offering and is the most valuable part of our products. The LithoData we have:

LithoData Premium - Lithodats mined data broken up into the following DataTypes

- Fission Track
- U-Th/He
- GeoChemistry
- U-Pb
- Therm. History
- Ar-Ar
- Vitrine
- Isotope
- Mineral Deposit Info

Classification: Product

LithoData is available in Local, Regional or Global purchasing packages. These packages are available through a licence agreement, with monthly or annual subscriptions. All data is delivered through the LithoSurfer platform. The customer must have at least a free Lithosurfer account.

Question: Where does it fall on the Matrix?

		Data:		
		Customer	Lithodat	Public
Work	Customer Work			
	Joint Work			
	Lithodat Work			
		Customer Site	Joint / Shared	Lithodat Site

LithoData™ - Marketplace

LithoData™ Marketplace is Lithodat's Data service that allows private companies (Retailers) to sell their data through Lithosurfer. All data must undergo a LithoClean service fee to incorporate the data into the system.

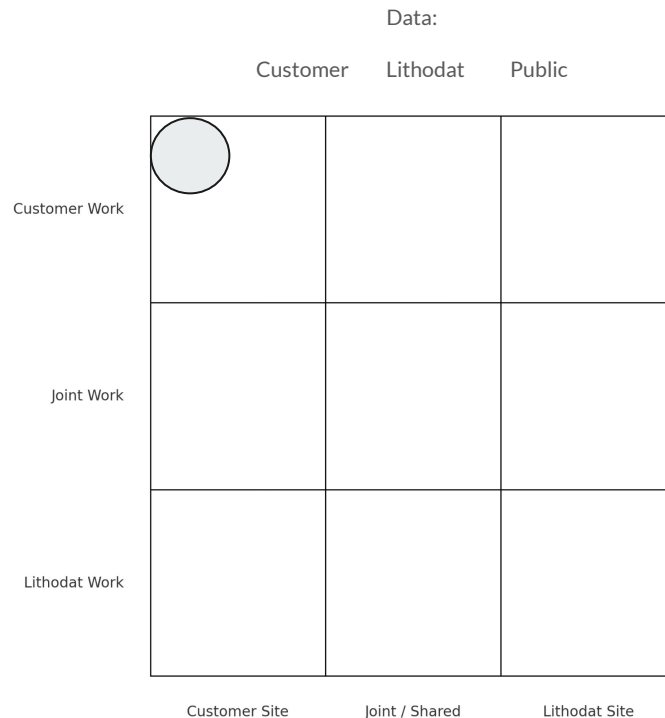
LithoData Market - Customers and Retailers can access packages of data for sale through the marketplace. The retailer can choose the price in which they want to sell the data and the transaction occurs through LithoSurfer. The data packages they can sell

- Fission Track
- U-Th/He
- GeoChemistry
- U-Pb
- Therm. History
- Ar-Ar
- Vitrine
- Isotope
- Mineral Deposit Info
- Geophysics Rasters
- Reports, Borelogs, Analysis

Classification: Service

LithoData Marketplace is separated into Packages and are sold as per the Retailers desire. These packages are available through a subscription or a perpetual licence. Lithodat takes a 30% cut (or negotiated fee dependant on the agreement with the retailer). Lithodat charges a LithoClean service fee to enable the highest quality data.

All data is viewed and delivered through the LithoSurfer platform. The customer must have at least a free Lithosurfer account.



Lithosurfer Free™

LithoSurfer Free is Lithodat's subscription product. It provides clients with all Litho Free data through the LithoSurfer platform. Users can explore all models and use basic tools within the system, but downloading is limited to prevent bulk export.

The Product Includes:

- Free signup into LithoSurfer
- Access to all LithoData Free packages
- Limited Tools
- Can upload unlimited public data
- Restricted private data upload
- Working within a secure, cloud-based environment maintained by Lithodat.
- Ability to purchase LithoData Premium

Classification: Product

LithoSurfer Free is a product: it is a standardised, repeatable subscription platform. All users receive the same access rights and tools, making it scalable and uniform across the customer base.

Customer-heavy, in their office — the client drives exploration and analysis through self-service. On the adapted matrix, LithoSurfer Basic sits firmly in the productised / customer-effort quadrant, representing a standardised, repeatable platform offering.

Lithodat Free uses only Publicly available or a Restricted dataset.

		Data:		
		Customer	LithoData	Public
Customer Work	Joint Work			
		Customer Site	Joint / Shared	Lithodat Site



Lithosurfer Pro™

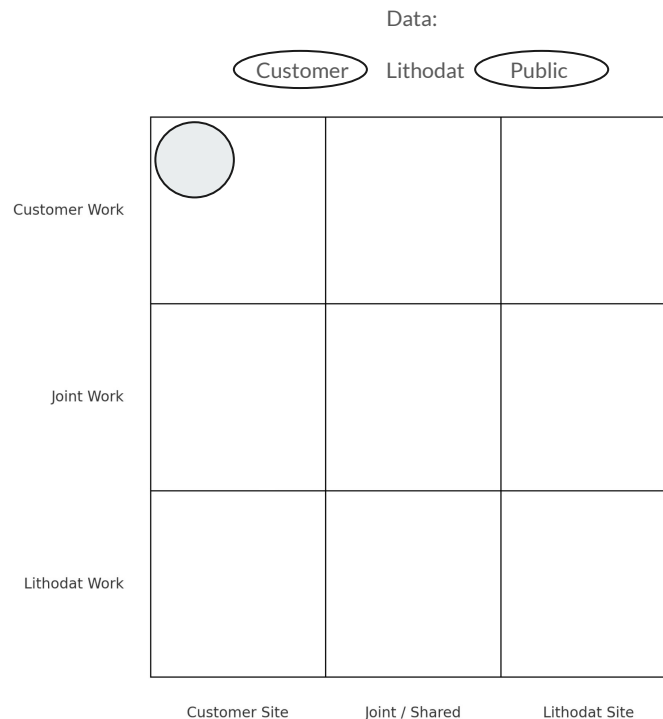
LithoSurfer Pro is Lithodat's professional subscription product. It costs 8000 AUD per seat per year. It provides clients with all the features of Lithodat Free. Pro subscribers also unlock advanced tools, increased private data holdings, expanded download capabilities, and access to the LithoSurfer API for integration with external workflows.

The features:

- Exploring all LithoData Free.
- Using premium analytical tools for filtering, modelling, and comparisons.
- Ability to upload semi-restricted amounts of private data.
- Can upload unlimited public data.
- Ability to purchase LithoData
- One seat per licence.
- Discounted LithoData Premium subscription
- Can be included in LithoData Premium subscriptions

Classification: Product - it is a repeatable subscription tier with a defined feature set and pricing structure. It has additional

Customer-heavy, in their office — the client drives exploration and analysis through self-service.



Lithosurfer Enterprise™

LithoSurfer Enterprise is Lithodat's advanced subscription product. In addition to all the public, Enterprise clients can also securely upload unlimited private datasets. This enables them to combine their internal data with Lithodat's holdings in a unified, private workspace while maintaining full control over visibility and access.

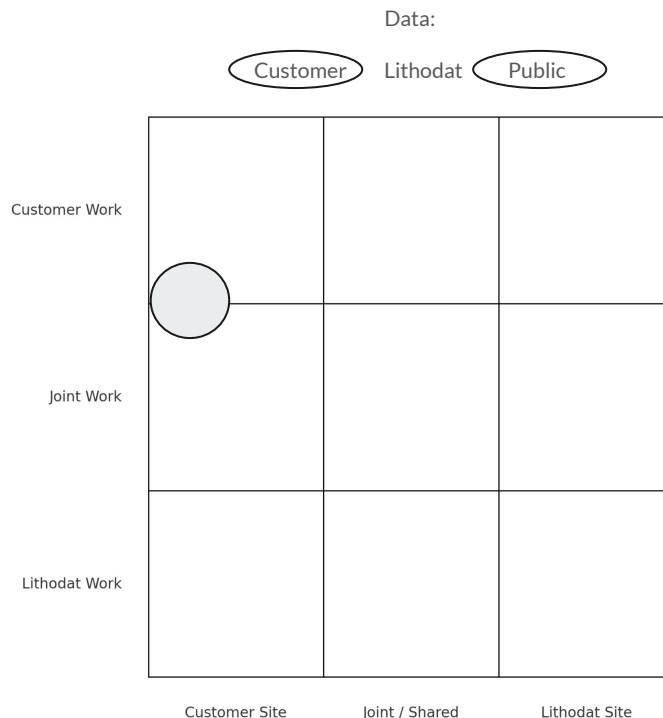
The Features:

- Customer Skins
- Accessing public in the platform.
- Using premium tools and API integrations for deep analysis.
- Uploading and managing proprietary client datasets in a secure, private environment
- High Security, SSO or private hosting.
- Discounted LithoData Premium
- Access to X amount of Clean™ Credits per year.
- Access to LithoBuild™ development
- Access to X amount of Mine™ Credits per year.
- Access to purchase LithoData - at a discounted price

Classification: Product - it is a defined subscription tier with advanced capabilities. It is scalable to any number of clients and represents the highest-value version of the LithoSurfer platform.

Customer-heavy, in their office — the client drives exploration, uploads their own data and analysis through self-service.

A mix of all three types of data we handle.



Combining Products and Services



Clean™ + Build™ = When we clean up customers data and then build them a platform or tools to view.

Mine™ + Clean™ = We find the data through Mine then Clean before transferring it to the Customer

LithoSurfer + Mine™ + Clean™ = Use our team to increase your exploration understanding of a region

Clean™ + LithoSurfer = Transition your whole dataset to a bespoke enterprise system, support mergers and acquisitions by unifying datasets, Transition entire government repositories into one system

LithoSurfer + Build™ = Include specific features, like adaptors or tools designed by the Customer



EarthBank™

EarthBank is Lithodat's AGN funded - free to use global geoscience data repository designed for students, researchers, and academics. It enables the storage, sharing, and discovery of both public and private research datasets, with full compliance to FAIR data standards.

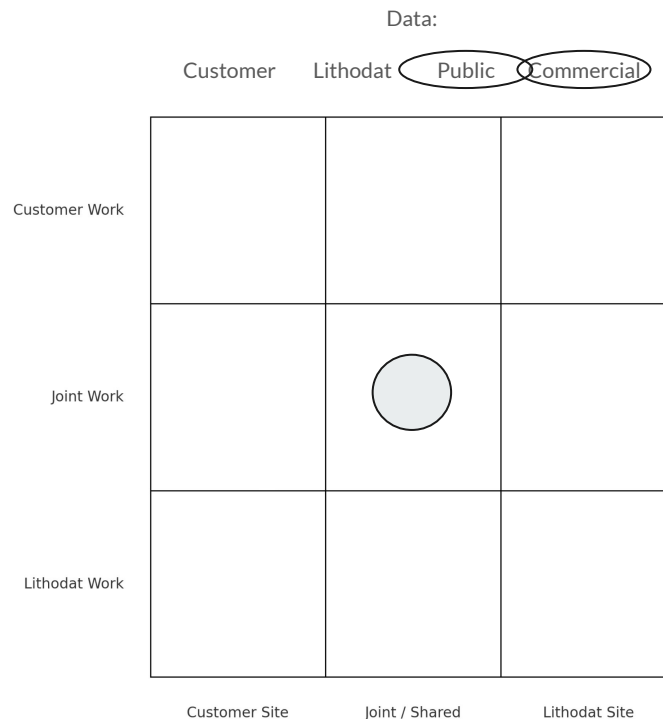
Features

- Free to use for academics - uploading and accessing datasets.
- Flexible sharing controls: datasets can be made public or kept private while research is in progress.
- Search and discovery tools for users to find and download datasets.
- Designed for Sharing and Collaboration

Classification: Product - it is a repeatable commercial platform where data can be uploaded.

Joint — Lithodat provides the platform and standards; researchers provide and maintain datasets.

This data is both public sometimes and commercial if held by the researcher.



Future Products



Other Financing Mechanisms

- Consulting and Training
- DOI and PID Minting
- Machine Learning and AI
- Tool Sandbox - Plugins and Adaptors
- Quality Systems - checking accuracy
- New Data Models (E.g. DERT)

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Clean™ + Lithosurfer = Transition your whole dataset to a bespoke enterprise system, support mergers and acquisitions by unifying datasets, Transition entire government repositories into one system

Lithosurfer + Build™ = Include specific features, like adaptors or tools designed by the Customer

Clean™ + Build™ + Market = Think Amira - Clean Products and a portal to sell through our system

Lithosurfer + Market = Free, Pro and Enterprise users have access to commercial data



Past and current jobs and their mapping to our products

Natural Resources Canada

AngloAmerican

BHP

AuScope EarthBank

CSIRO Isotopes.au

LITH  DAT

LITH  DAT

Lithosurfer Free™



LithoSurfer Free is Lithodat's entry-level subscription product. It provides clients with all LithoFree data through the LithoSurfer platform. Users can explore all models and use basic tools within the system, but downloading is limited to prevent bulk export.

The Product Includes:

- Free signup into LithoSurfer
- Access to all LithoData Free packages
- Limited Tools
- Can upload unlimited public data
- No private data upload
- Working within a secure, cloud-based environment maintained by Lithodat.
- Ability to purchase LithoData Premium

Classification: Product

LithoSurfer Free is a product: it is a standardised, repeatable subscription platform. All users receive the same access rights and tools, making it scalable and uniform across the customer base. the

Question: Where does it fall on the Matrix?

Client Data? Public Data? Academic Data?

Data:

Customer

Lithodat

Public

Customer Work

Joint Work

Lithodat Work

Customer Site

Joint / Shared

Lithodat Site



Lithosurfer Pro™

LithoSurfer Pro is Lithodat's professional subscription product. It provides clients with access to purchase access to Lithodat Free **plus** Lithodat's additional geoscience data holdings (e.g. thermochronology, vitrinite reflectance, and other global datasets). Pro subscribers also unlock advanced tools, expanded download capabilities, and access to the LithoSurfer API for integration with external workflows.

The process involves:

- Exploring both public and proprietary Lithodat datasets within the secure platform.
- Using premium analytical tools for filtering, modelling, and comparisons.
- Ability to upload restricted amounts of private data
- Can upload unlimited public data.
- Ability to purchase LithoData
- One seat per licence.

Classification: Product - it is a repeatable subscription tier with a defined feature set and pricing structure.

Question: Where does it fall on the Matrix?

Client Data? Public Data? Academic Data?

Data:

Customer

Lithodat

Public

Customer Work

Joint Work

Lithodat Work

Customer Site

Joint / Shared

Lithodat Site



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LithoSurfer Enterprise is Lithodat's advanced subscription product. In addition to all the public and proprietary Lithodat datasets available in Pro, Enterprise clients can also securely upload and manage their own private datasets. This enables them to combine their internal data with Lithodat's holdings in a unified, private workspace while maintaining full control over visibility and access.

The Features:

- Customer Skins
- Accessing public in the platform.
- Using premium tools and API integrations for deep analysis.
- Uploading and managing proprietary client datasets in a secure, private environment - still restricted to XXX amount of datapoints.
- Highly Security
- Access to X amount of Clean™ Credits per year.
- Access to LithoBuild™ development
- Access to X amount of Mine™ Credits per year.
- Access to purchase LithoData - at a discounted price

Classification: Product - it is a repeatable subscription tier with a defined feature set and pricing structure.

Question: Where does it fall on the Matrix?

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		Data:			
		Customer	Lithodat	Public	
Customer Work	Joint Work				
Lithodat Work					
		Customer Site	Joint / Shared	Lithodat Site	



Market™



Market is Lithodat’s commercial data marketplace inside the LithoSurfer platform. It enables approved external sellers — academic groups, consortia, labs, or companies — to list and sell their cleaned, standardised datasets directly to LithoSurfer users. Lithodat manages the ingestion, hosting, transaction, and delivery, while sellers retain ownership and control over pricing and licensing.

The service creates a discoverable environment for external data allowing Lithodat clients to access commercial datasets alongside public and proprietary data. A rating system ensures data quality is transparent.

Classification: Product - it is a repeatable commercial platform where data can be transacted. Lithodat facilitates the cleaning of the data but does not own the listed datasets. Instead, it takes a transaction share while sellers maintain ownership.

Question: Where does it fall on the Matrix?

Client Data? Public Data? Academic Data?

		Data:		
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Customer Work	Joint Work	Lithodat Work		
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Features

- Free to use for academics - uploading and accessing datasets.
- Flexible sharing controls: datasets can be made public or kept private while research is in progress.
- Search and discovery tools for users to find and download datasets.
- Integration pathway into LithoSurfer for curated, long-term use.

Classification: Product - it is a repeatable commercial platform where data can be uploaded.

Question: Where does it fall on the Matrix?

Client Data? Public Data? Academic Data?

Data:

Customer Lithodat Public Commercial

Customer Work			
Joint Work			
Lithodat Work			
	Customer Site	Joint / Shared	Lithodat Site



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- Identifying inconsistencies in fields, units, naming, and metadata.
- Converting them into a structured, relational database format.

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- Building adaptors and tools inside Lithosurfer requested by customers.
- Developing whole new platforms or extending existing ones for government surveys, academia, or industry. (ISOtopes)

Classification: Service

Lithodat's experts and processes deliver a result based on the customer's unique raw data. The output may be a product, but BUILD is specific to the service we provide.

Question: Where does it fall on the Matrix?

Client Data? Public Data? Academic Data?

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Work	Customer Work			
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	Lithodat Work			
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- Delivering the results as a structured, cleaned dataset ready for use.

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Client Data? Public Data? Academic Data?

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Joint Work				
Lithodat Work				
		Customer Site	Joint / Shared	Lithodat Site



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The service creates a discoverable environment for external data allowing Lithodat clients to access commercial datasets alongside public and proprietary data. A rating system ensures data quality is transparent.

Classification: Product - it is a repeatable commercial platform where data can be transacted. Lithodat facilitates the cleaning of the data but does not own the listed datasets. Instead, it takes a transaction share while sellers maintain ownership.

Question: Joint — Lithodat provides the infrastructure and process, sellers provide the data. Lithodat manages the queries and interaction with the buyers, the customers provide the data

Client Data? Public Data? Academic Data?

None of the Above - this is a new category - commercial data. Data that is owned neither by us nor the end user. Best of all it doesn't need to be geospatial data we sell: **Journals, Reports, Site Photos, Geophysics...**

Data:

	Customer	Lithodat	Public	Commercial
Customer Work				
Joint Work				
Lithodat Work				
	Customer Site	Joint / Shared	Lithodat Site	

LithoData™

LithoData™ is Lithodat's Data as a Service (DaaS) offering and is the most scalable part of our products. Lithodat has access to our own LithoMined datasets that are separate from the from various regions across the world expert teams harvest the public domain (academic journals, government repositories, survey reports) and systematically extracting the data and cleaning it and putting it in data packages. The process involves:

- Searching across diverse public and semi-public sources.
- Extracting relevant datasets and metadata.
- Delivering the results as a structured, cleaned datapackageready for use.

Classification: Service

DATA is a data extraction-only activity: Lithodat's team uses its technology, resources, and expertise to source and assemble data. The output is a consolidated in data packages, that will then be sold as a subscription.

Question: Where does it fall on the Matrix?

Commercial Data

Data:

Customer Lithodat Public

	Customer Site	Joint / Shared	Lithodat Site
Customer Work			
Joint Work			
Lithodat Work			



Lithodat Utopia - FK

Platform: Become the Google of Geosciences! Global leader/place to go for standardised, AI/ML ready geoscientific sample based data (easy to Find, access and use). Easy to use and fully integrated tools and analytics helping to find the data for the users as fast and easy as they can without the need of external programs.

Customers:

- Any geoscience, earth samples industries (Mining, Oil&Gas, Geothermal, Energy, Construction, Agriculture, Health (Contam))
- Labs (drop, publish, sell data)
- Researcher (find, upload, publish, share data)
- Governments (publish, share, sell?)
- AI (pay subscription)



Lithodat Utopia - FK

Culture: International mindset, open, fair and very important a no-asshole policy! Innovative, exchange of ideas, -> always in the forefront of tech, latests tools, AI etc...

Employees:

- Different offices - different products, teams
- Subject matter experts (SME) per topic, product, data type, theme
- Product owners part of the SME teams
- Management

Assets:

- Platform and tools
- DATA, DATA and DATA
- IP (patents?), trademarks, copyright
- Real estate...? Offices etc?



Lithodat Utopia - FK

Turnover in 2, 5 and 10years:

2y:	>5M	20 - 30 people
5y:	>25M	500 - 100 people
10y:	>250M	300 - 800 people